

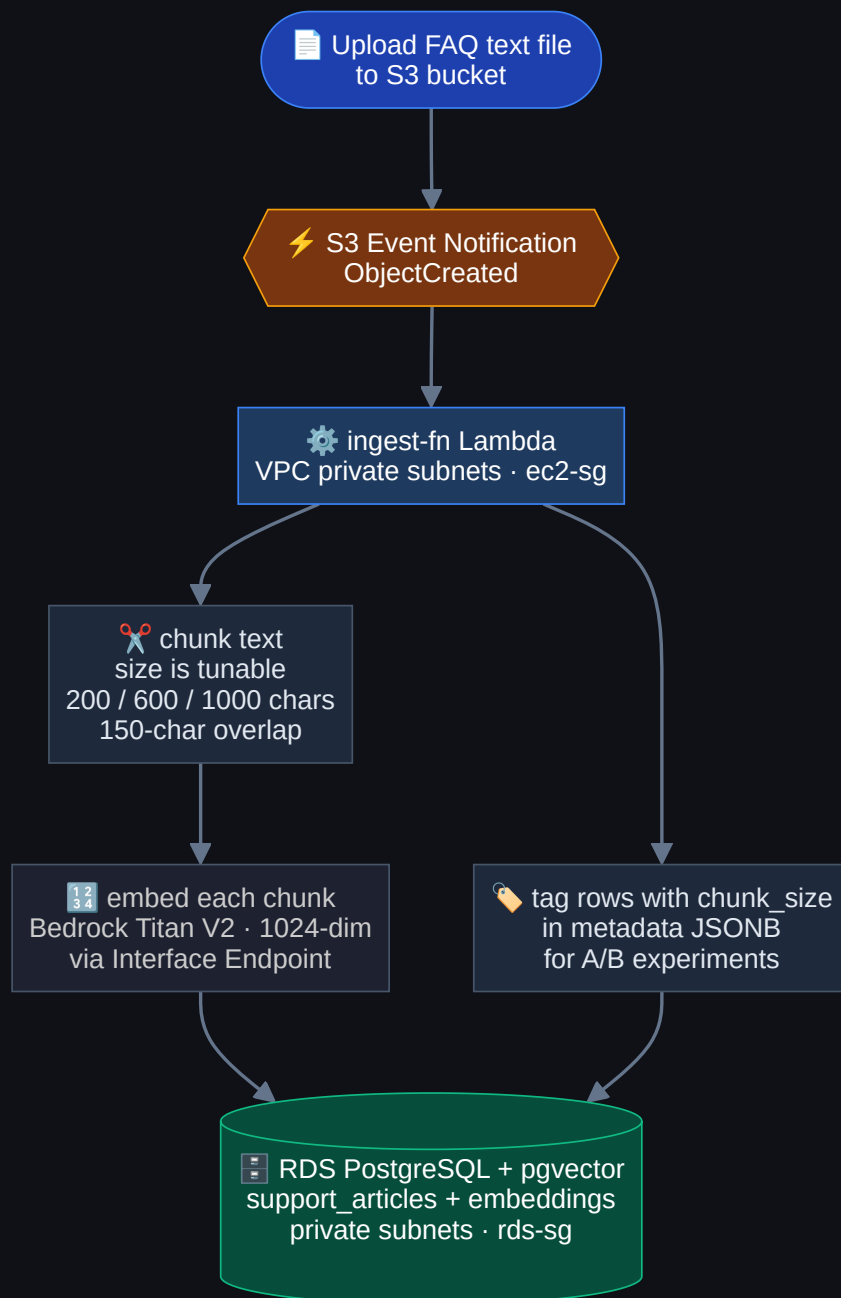


Smart Help Desk — RAG Optimizer

End-to-End Flowcharts — Hybrid Architecture (Lambda Ingestion + EC2/ALB Query) · Tuning Lab · Solutions Architect Prep

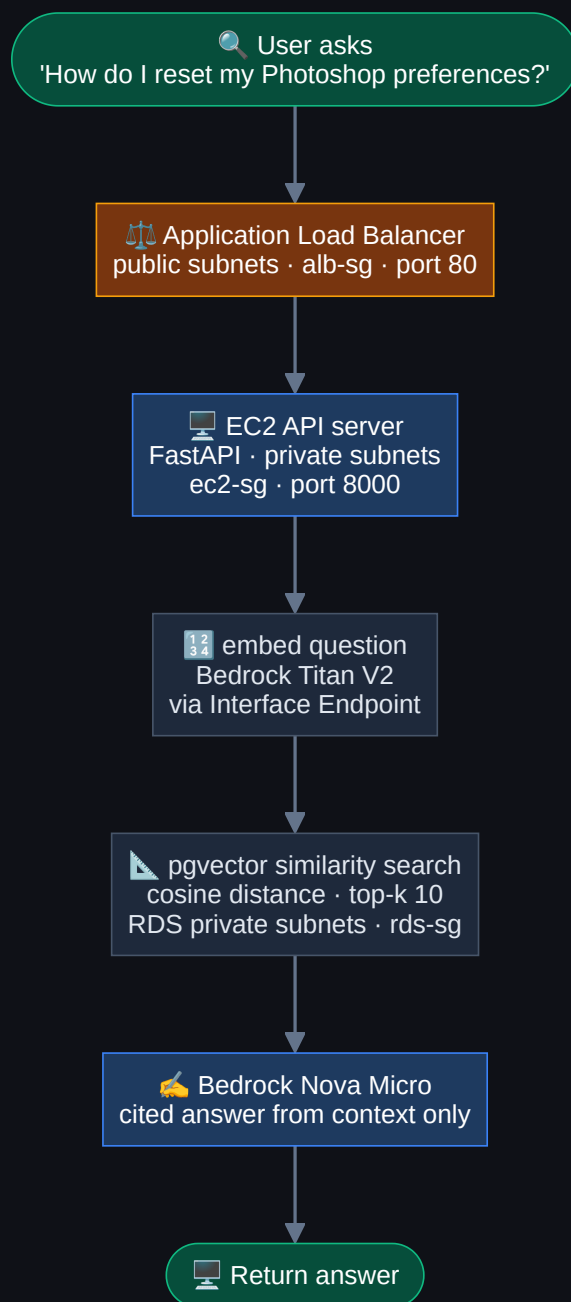
● Pipeline A — Ingestion (The Indexer)

You drop an FAQ file into S3. That event automatically triggers the ingest - fn Lambda (private subnets, ec2 - sg) to read the file through the free S3 Gateway Endpoint, chunk the text, embed it with Bedrock via the Interface Endpoint, and store it in RDS pgvector.



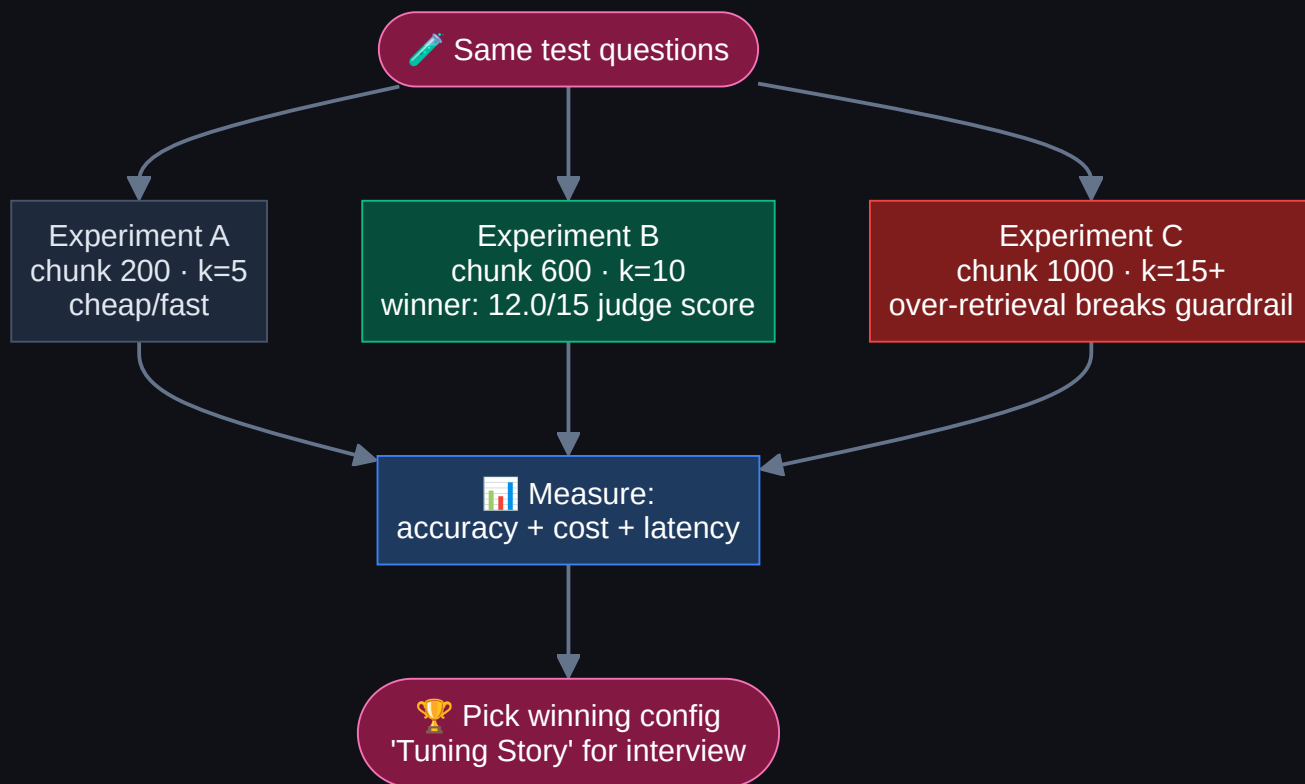
● Pipeline B — Query (EC2 + ALB Answer Engine)

A user's question comes in through the Internet Gateway to the Application Load Balancer (public subnets), which forwards it to the EC2 API server (private subnets). EC2 embeds the question via the Bedrock Interface Endpoint, searches pgvector, then asks Nova Micro for a cited answer.



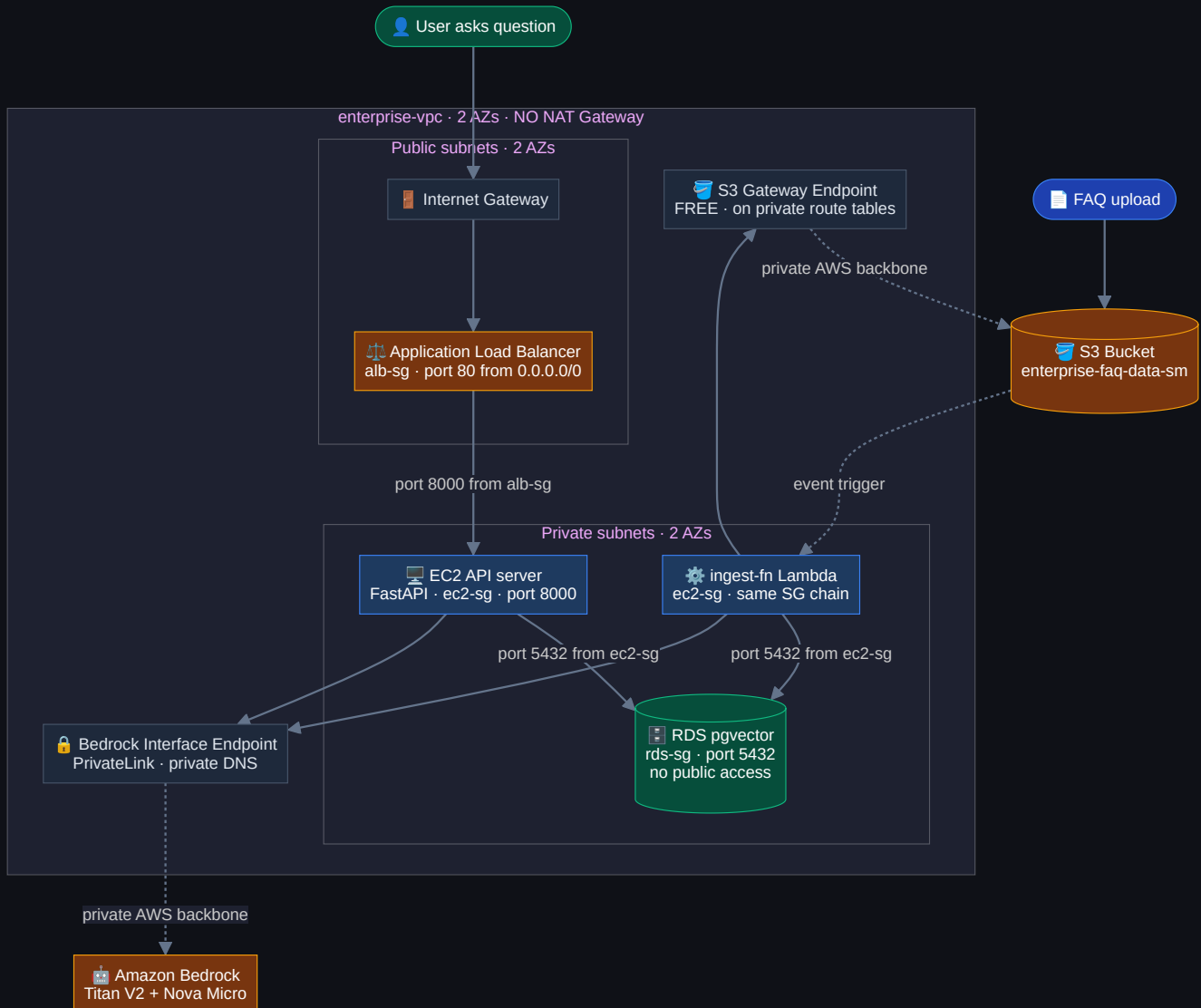
♥ The Tuning Lab (What makes this "Proper")

The differentiator. You run the SAME questions through different configurations, measure the results, and pick the best settings — giving you real metrics for your resume.



● Infrastructure View — Hybrid VPC (Lambda + EC2/ALB)

The full picture. Ingestion is serverless (S3 → Lambda). The query API is enterprise (Internet Gateway → ALB in public subnets → EC2 in private subnets). NO NAT Gateway: private compute reaches S3 over a **free S3 Gateway Endpoint** and reaches Bedrock over a **Bedrock Interface Endpoint (PrivateLink)**. Three-tier security group chain: alb-sg (80 from the world) → ec2-sg (8000 from alb-sg) → rds-sg (5432 from ec2-sg). The Lambda reuses ec2-sg, so it inherits the same chain.



⚡ Sequence Flash Cards

● Ingestion — say it in order:

Upload to S3 → S3 Event Trigger → ingest-fn Lambda (private subnets · ec2-sg) → [read file via S3 Gateway Endpoint · free] → Chunk (600 · 150 overlap) → Embed – Titan V2 via Bedrock Interface Endpoint → [RDS pgvector (rds-sg · 5432) + metadata tag]

● Query — say it in order:

Question → Internet Gateway → ALB (public subnets · alb-sg · 80) → EC2 API (private subnets · ec2-sg · 8000) → Embed – Titan V2 via Bedrock Interface Endpoint → [pgvector top-k 10 (rds-sg · 5432)] → Nova Micro Synthesis → Cited Answer

♥ Tuning Lab — say it in order:

Same questions → [chunk 200–1000] → [k=5 vs k=10 vs k=15+] → LLM-as-a-Judge score → Pick winner: chunk 600, K=10

1 2
3 4

Key Numbers to Memorize

600

Winning chunk size (locked in production)

10

Winning Top-K (locked in production)

150

Char overlap between chunks

1024

Titan V2 embedding dimensions

100% →

50%

Adversarial block rate at K=10 vs K≥15

\$0

NAT Gateway (avoided! saves ~\$32/mo) — free S3 Gateway Endpoint instead

3

Security group chain: alb-sg → ec2-sg → rds-sg

2

Bedrock models: Titan + Nova Micro

1

Database: RDS pgvector

\$10

Budget alarm (total spend ~\$1-5)

2

Days to build



Well-Architected Pillars (one-liners)

Operational Excellence

CloudWatch logs

Security

Private-subnet DB · least-privilege IAM · SG firewall

Reliability

Managed RDS · event-driven retries · idempotent writes

Performance

ivfflat index · tuned chunk size · tuned top-k

Cost Optimization

Free-tier · NO NAT Gateway · top-k tuning saves tokens

Sustainability

Serverless scales to zero — no idle compute